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Q1

$T(n) = T(n/2) + O(1)$. Unrolling: after k halvings the array has $n/2^k$ elements; we stop when $n/2^k = 1$, i.e. $k = \log_2 n$. Each level does $O(1)$ work, so $T(n) = O(\log n)$. Worst case: the element is absent, and we probe until the range is empty — still $\log_2 n + 1$ probes.

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Q2

Pick the shortest interval first — the shortest one blocks the fewest others, so by induction the shortest-first greedy is optimal. QED.

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Q3

Let $L[i]$ = length of the LIS ending exactly at $a[i]$.

$L[i] = 1 + \max\{ L[j] : j < i, a[j] < a[i] \}$ (or 1 if none). Answer $\max L[i]$.

Two nested loops give $O(n^2)$. Refinement: keep $tails[k]$ = smallest tail of an increasing subsequence of length $k+1$; binary-search each element: $O(n \log n)$.

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Q4

False: BFS can take a longer path if it enqueues vertices in a bad order,
e.g. a diamond graph where the right branch is explored first.